

llmXive Automated Review of FastContext: Training Efficient Repository Explorer for Coding Agents

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer’s exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer’s section also names the underlying model that produced it.

This paper’s panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: [qwen.qwen3.5-122b](#)

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and the validity of their supporting citations.

Citation Mismatch in Baselines: In the “Standalone Exploration Evaluation Protocol” section (e000), the authors list “OpenHands-Bash” as a baseline and cite it with `\cite{openhands, codescout}`. While `\cite{openhands}` correctly references the OpenHands framework, `\cite{codescout}` refers to “CodeScout: An Effective Recipe for Reinforcement Learning of Code Search Agents.” There is no logical connection in the provided text or bibliography that suggests the CodeScout paper defines or implements the “OpenHands-Bash” baseline. This appears to be a citation error where the wrong key was included, potentially misleading readers about the source of the baseline implementation. The authors should verify the correct citation for the OpenHands-Bash baseline or remove the extraneous `codescout` key if it does not apply to that specific baseline.

Unsubstantiated Data Provenance: The “SFT Data Construction” section (e000) claims the training corpus of 2,954 examples is “generated from Sonnet 4.6 exploration traces.” While the paper cites `\cite{qwen3}` for the Qwen3 backbones, it provides no citation, link, or reference for “Sonnet 4.6” or the methodology used to generate these traces. Given that Sonnet is a proprietary model (Anthropic), the specific version “4.6” and the process of extracting “exploration traces” from it are critical details for reproducibility. The lack of a reference for the data source makes this claim difficult to verify and weakens the scientific rigor of the data construction section.

Missing Cost Calculation Details: In the “Runtime Integration and Token Accounting” section (e000), the authors present specific cost figures: a reduction from 282.47 to 208.92, saving \$69.03. While the paper explicitly cites the Fireworks serverless pricing (\$0.20/1M tokens) for the *explorer* cost calculation, it fails to state the specific API pricing tier or cost-per-token rate used for the *main agent* (GPT-5.4) to derive the 282.47 and 208.92 figures. Without this information, the reader cannot independently verify the cost savings claim. The text mentions “provider-recorded GPT-5.4 API cost” in a figure caption, but the specific rate or total token count used for the main agent in the text description is not explicitly linked to a verifiable price point in the text itself.

Action items.

- **[writing]** The review focuses on the accuracy of factual claims and the validity of their supporting citations. Citation Mismatch in Baselines: In the “Standalone Exploration Evaluation Protocol” section (e000), the authors list “OpenHands-Bash” as a baseline and cite it with `\cite{openhands, codescout}`. While `\cite{openhands}` correctly references the OpenHands framework, `\cite{codescout}` refers to “CodeScout: An Effective Recipe for Reinforcement Learning of Code Search Agents.” There is no logical connect

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

The figure effectively visualizes the tradeoff but suffers from a non-linear x-axis that distorts the data representation and a fragmented legend that complicates symbol interpretation. Additionally, the caption contains a grammatical error that obscures the subject of the claim.

Figure 2

Figure 2 effectively visualizes the cost reduction but lacks statistical context (N, error bars) to validate the ‘Estimated API cost’ claim, and the x-axis labels are cluttered.

Figure 3

The figure presents stacked bar charts for tool turns and token shares, but the caption’s reference to ‘Left’ and ‘Right’ panels does not match the vertical layout shown. Additionally, the token breakdown lacks a value for the ‘Other’ category, undermining the caption’s claim about dominance of specific actions.

Figure 4

The figure provides a clear visual overview of the agent architecture, but the caption is critically broken with missing nouns and grammatical errors that obscure the subject and the specific example shown.

Figure 5

The figure effectively visualizes the shift in token distributions, but the quantitative annotations (red arrows and percentages) lack a definition in the caption, and the y-axis labels are missing from the last three subplots.

Figure 6

The figure effectively visualizes token reduction across benchmarks, but the Sankey diagram labels are inconsistent with the absolute token counts provided, and the ‘FastContext’ category is missing from the right-side breakdown labels. Additionally, the caption has a grammatical error omitting the subject of the main claim.

Figure 7

The figure fails to match its caption, which promises a two-panel layout (SFT loss and RL reward) but only displays the SFT loss plot. Additionally, the x-axis label is poorly positioned,

overlapping the data curves.

Action items.

- **[writing]** Figure 1: The caption contains a grammatical error (‘where shifts coding agents’) and omits the subject (likely ‘FastContext’), making the claim unclear.
- **[science]** Figure 1: The x-axis uses a non-linear, irregular scale (e.g., 200k to 300k is wider than 1M to 1.5M) which visually distorts the slope of the tradeoff lines and misrepresents the magnitude of token savings.
- **[writing]** Figure 1: The legend is split into two separate blocks (‘Main Model’ and ‘FastContext Model’) with inconsistent formatting, making it difficult to map the combined symbols (e.g., circle with square) to their specific components.
- **[science]** Figure 2: The y-axis label ‘Estimated API cost (USD)’ and the bar values (282.47,208.92) imply a total cost per benchmark run, but the caption describes these as ‘provider-recorded GPT-5.4 API cost’ without specifying if this is a mean, median, or total across the SWE-bench Multilingual suite. Without a sample size (N) or error bars, the magnitude of these costs is ambiguous and difficult to interpret as a standard metric.
- **[writing]** Figure 2: The x-axis label ‘4B-RL subagent’ is visually crowded and the text is split across three lines, reducing readability. Additionally, the annotation ‘2.2% of augmented main cost’ is placed directly above the bar but lacks a clear visual connector (like a bracket) to the ‘augmented main cost’ bar, requiring the reader to infer the comparison.
- **[science]** Figure 3: The caption claims the figure shows ‘reading and searching dominate... prompt-token usage’, but the bottom panel (‘Total tokens’) displays percentages (32.4%, 14.1%, etc.) that sum to ~93% with the ‘Other’ category unlabelled. It is unclear if these are shares of total tokens or just the visible categories, and the ‘Other’ slice is not quantified, making the claim of dominance difficult to verify.
- **[writing]** Figure 3: The caption refers to ‘Left’ and ‘Right’ panels, but the figure displays two stacked horizontal bar charts (‘Turns’ and ‘Total tokens’) without clear left/right separation or labels indicating which part corresponds to which claim in the caption.
- **[fatal]** Figure 4 caption contains multiple grammatical errors and missing nouns (e.g., ‘Overview of .’, ‘delegates repository exploration to and receives’), making it impossible to identify the subject of the figure.
- **[science]** Figure 4: The ‘Fast Context Agent’ box contains a specific example (‘Query(hugo-12204)’) and code snippets that are not explained in the caption, making the specific workflow difficult to interpret without external context.
- **[science]** Figure 5: The red downward arrows and percentage values (e.g., ‘↓ 17.1%’) are not defined in the caption or legend. It is unclear if these represent the shift in mean, median, or a specific quantile, making the quantitative claim ambiguous.
- **[writing]** Figure 5: The y-axis label ‘Frequency’ is present only on the first subplot; the remaining three subplots lack y-axis labels, which is inconsistent formatting.

- **[science]** Figure 6: The Sankey diagram labels ‘File Reading’ (43%) and ‘Code Search’ (19%) on the left side sum to 62%, but the total token count is 818k. The right side labels ‘File Reading’ (40%) and ‘Code Search’ (18%) sum to 58% of 701k. The percentages do not align with the absolute token counts shown below the bars, and the ‘FastContext’ category (purple in legend) is missing from the breakdown labels on the right side despite being part of the total.
- **[writing]** Figure 6: The caption contains a grammatical error and missing subject: ‘substantially reduces main-agent context consumption’ lacks the subject ‘FastContext’ or ‘FC-4B-RL’.
- **[science]** Figure 7: The caption describes a two-panel figure (‘Left: ... Right: ...’), but the rendered image shows only a single plot titled ‘SFT Loss’. The ‘Right’ panel showing reinforcement-learning reward is missing.
- **[writing]** Figure 7: The x-axis label ‘train/step’ is placed directly on top of the data lines near x=130, making it illegible and obscuring the plot area.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript exhibits significant jargon overuse, particularly in the Introduction and Method sections, which hinders accessibility for non-specialist readers. The most critical issue is the undefined macro `\approach` (e.g., in the Introduction: “`\approach` is a read-only explorer”), which appears to be a placeholder for the system name “FastContext” but is never defined, rendering the text unintelligible. Additionally, acronyms such as SFT, RL, GRPO, SWE-bench, F1, API, CLI, JSONL, and LLM are used frequently without being spelled out at their first occurrence. While these are standard in the field, the paper’s goal of broad accessibility requires that they be defined. The use of model-specific names like “Sonnet 4.6” and “Qwen3” without context also assumes prior knowledge. The text should be revised to define all acronyms and technical terms upon first use, and the `\approach` macro should be replaced with the full system name or defined clearly.

Action items.

- **[writing]** The manuscript exhibits significant jargon overuse, particularly in the Introduction and Method sections, which hinders accessibility for non-specialist readers. The most critical issue is the undefined macro `\approach` (e.g., in the Introduction: “`\approach` is a read-only explorer”), which appears to be a placeholder for the system name “FastContext” but is never defined, rendering the text unintelligible. Additionally, acronyms such as SFT, RL, GRPO, SWE-bench, F1, API, CLI, JSONL, and LLM are

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

“The logical consistency of the paper is generally sound, with clear causal links between the proposed method (decoupled exploration) and the observed outcomes (token reduction). However, there are minor ambiguities in the presentation of cost accounting and the evaluation protocol that require clarification to ensure the conclusions follow strictly from the premises.\n\nFirst, in the ‘Runtime Integration and Token Accounting’ section, the paper presents a cost breakdown where the total system cost is 213.44(208.92 for the main agent + \$4.52 for the explorer). The text states the explorer accounts for ‘2.1%’ of the total. While the arithmetic ($4.52 / 213.44 = 2.12\%$) is correct, the phrasing creates a slight logical ambiguity. It is unclear if the ‘2.1%’ figure is intended to highlight the explorer’s overhead relative to the **total** system cost or relative to the **original** main-agent cost (\$282.47). If the latter, the percentage would be $\sim 1.6\%$. Clarifying the denominator used for this percentage is necessary to ensure the claim of ‘negligible overhead’ is logically supported by the specific metric cited.\n\nSecond, the ‘Standalone Exploration Evaluation Protocol’ (Appendix) describes a setup where the ‘reference patch is hidden’ from the explorer, yet the evaluation relies on ‘patch-derived ground truth’. While this is standard for evaluation, the text could be more precise to avoid the logical impression that the ground truth is used during the inference phase. Explicitly stating that the ground truth is *only* used for the post-hoc calculation of Precision/Recall/F1, and not as an input or reward signal during the exploration phase, would strengthen the logical rigor of the experimental design description.\n\nFinally, the relationship between the SFT data and the RL reward function warrants a brief clarification. The SFT corpus includes a specific split for ‘parallel_toolcalls’ (990 examples), suggesting the model is already trained to issue parallel calls. The RL reward function r_{parallel} explicitly rewards parallelism ($3 < p_{\text{max}} \leq 6$). The paper should briefly explain the causal role of this reward: is it intended to *reinforce* a behavior that SFT only partially learned, or to *optimize* the quality/efficiency of parallel calls that SFT might have learned sub-optimally? Without this distinction, the necessity of the RL step for inducing parallelism is logically ambiguous, even if the results show improvement.”

Action items.

- **[writing]** “The logical consistency of the paper is generally sound, with clear causal links between the proposed method (decoupled exploration) and the observed outcomes (token reduction). However, there are minor ambiguities in the presentation of cost accounting and the evaluation protocol that require clarification to ensure the conclusions follow strictly from the premises.\n\nFirst, in the ‘Runtime Integration and Token Accounting’ section, the paper presents a cost breakdown where the total system c

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several strong claims regarding cost savings and performance gains that require tighter qualification to avoid overreach.

First, in the “Runtime Integration and Token Accounting” section, the authors calculate a specific cost saving of **\$69.03** and state the explorer accounts for **2.1%** of the total cost. This calculation is derived by applying a “Conservative API pricing” model (\$0.20/1M tokens) to the 4B-RL explorer’s token usage. However, the text immediately notes that “In deployment the explorer runs locally, so no API cost is incurred.” Claiming a specific dollar savings based on a hypothetical API cost for a component that is intended to be free (local) is an over-extrapolation. The savings are only real if the alternative is running the *entire* system via a paid API, but the comparison mixes a local component cost (simulated) with a main-agent API cost. The authors should rephrase this to clarify that the “savings” are relative to a hypothetical scenario where the explorer is also a paid API service, rather than a realized deployment benefit.

Second, the Conclusion and Introduction claim accuracy gains “up to 5.5%” and token reductions “up to 60%”. While these numbers appear in the results (e.g., GPT-5.4 on SWE-bench Pro for accuracy, GPT-5.4 on SWE-QA for tokens), the phrasing suggests these are generalizable upper bounds for the method. The paper should explicitly state that these maximums are specific to certain benchmark/agent combinations (e.g., “up to 5.5% on SWE-bench Pro with GPT-5.4”) to prevent readers from assuming these gains apply to all evaluated settings, particularly given the variance seen in the “End-to-End Case Studies” where one instance showed increased token usage.

Finally, the claim that the “4B-RL explorer often matches or exceeds 30B-SFT” is supported by the file-level F1 scores in Table 1 (73.71 vs 73.71/72.13). However, the text does not explicitly limit this claim to file-level performance. In module-level and function-level metrics, the 30B-SFT model often retains an advantage. Broadening the claim to “matches or exceeds 30B-SFT” without qualifying the granularity (file-level) risks overgeneralizing the efficiency gains of the smaller model.

Action items.

- **[writing]** The claim that the explorer ‘accounts for 2.1% and the system saves \$69.03’ (Section: Runtime Integration) relies on a counterfactual API pricing model (\$0.20/M tokens) for the 4B-RL explorer. Since the paper states the explorer runs locally in deployment, extrapolating a specific dollar savings based on serverless API costs is an over-claim of economic impact. Clarify that the savings are relative to a hypothetical API-based baseline, not a realized deployment cost reduction.
- **[writing]** The conclusion states the method improves success ‘up to 5.5%’ and cuts tokens ‘up to 60%’. The text cites ‘GPT-5.4 on SWE-bench Pro’ for the former and ‘GPT-5.4 on

SWE-QA’ for the latter. Ensure these specific maximums are explicitly highlighted in the main text or abstract to prevent readers from inferring these gains apply universally across all benchmarks and agents.

- **[writing]** The paper claims the 4B-RL explorer ‘often matches or exceeds 30B-SFT’ (Section: Experiments). While Table 1 shows high performance, the text does not explicitly qualify that this comparison holds primarily for file-level F1 or specific benchmarks. Broaden the claim to specify the granularity (e.g., ‘matches 30B-SFT on file-level F1’) to avoid over-generalizing performance across all metrics.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript addresses safety and ethics with a dedicated section (e001/e002), correctly identifying that the read-only explorer could indirectly facilitate the creation of insecure code if the main agent acts on flawed citations. However, the discussion remains high-level. The authors should expand on the “over-reliance” risk by specifying if the system architecture enforces any verification steps (e.g., mandatory test runs) before the main agent accepts the explorer’s citations as ground truth.

Regarding data privacy, the paper notes the use of public repository data but does not explicitly detail the handling of the “Sonnet 4.6” traces used for SFT data construction. Given the potential for proprietary code leakage in such traces, a brief confirmation of the licensing and filtering process for these specific training examples is necessary to rule out copyright or confidentiality violations.

Finally, while the authors state that no new human-subject data was collected, the evaluation relies on a specific subset of SWE-bench Pro (200 instances listed in the appendix). A short confirmation that this subset was screened for sensitive Personally Identifiable Information (PII) or offensive content—beyond what is standard in the benchmark—would strengthen the ethics statement, ensuring that the release of evaluation results does not inadvertently highlight or propagate such content.

Action items.

- **[writing]** The ‘Potential Risks’ section (e001/e002) acknowledges that the tool may enable agents to introduce bugs or insecure changes but lacks specific mitigation strategies for the ‘over-reliance’ risk. Explicitly state how the system prevents the main agent from blindly executing code based solely on the explorer’s citations without verification steps.
- **[writing]** The paper mentions using ‘Sonnet 4.6’ traces for SFT data construction (e000, e001). Clarify the data provenance and licensing status of these traces to ensure no proprietary or non-public code was inadvertently included in the training set, which could pose legal or

privacy risks.

- **[writing]** While the paper states no new human-subject data was collected, the evaluation uses SWE-bench instances derived from real GitHub issues. Add a brief statement confirming that the specific 200-instance subset of SWE-bench Pro (listed in e002/e003) does not contain sensitive PII or offensive content that was not already publicly visible, or describe the filtering process used.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The scientific evidence supporting the central claims of FastContext is generally robust in terms of experimental design but relies on small sample sizes for the core training phases, which introduces potential variance risks.

Training Data and Sample Size: The Reinforcement Learning (RL) phase utilizes a corpus of only **400 prompts** across **395 repositories** (Section `app:rl-data`). While the authors report significant improvements in file-level F1 (73.71) and end-to-end success rates, the small sample size for the RL optimization makes the results susceptible to high variance. The paper does not report standard deviations, confidence intervals, or results from multiple random seeds for the RL training. Without these, it is difficult to determine if the observed gains are robust or if they result from the specific selection of the 400 prompts. Similarly, the SFT dataset of **2,954 examples** (Section `app:sft-data`) is relatively small for training a specialized agent, raising concerns about overfitting to the specific trajectory styles of the Sonnet 4.6 teacher model.

Evaluation Scope: The end-to-end evaluation (Table `tab:e2e_main`) primarily highlights results using **GPT-5.4** as the main agent. While the paper claims the method is a general solution for “coding agents,” the evidence is heavily weighted toward a single, high-capacity main model. To substantiate the claim that the explorer is broadly effective, the authors should ideally demonstrate that the token savings and success rate improvements persist when paired with different main agent architectures or smaller models. The current evidence suggests the method works well with GPT-5.4 but leaves the generalizability to other agent configurations as an assumption rather than a demonstrated fact.

Statistical Rigor: The paper reports point estimates for success rates and token counts (e.g., “5.5% accuracy gain,” “60% token reduction”) without accompanying statistical significance tests (e.g., t-tests or bootstrap p-values) given the fixed benchmark subsets (e.g., the 200-instance SWE-bench Pro subset). While the effect sizes appear large, the lack of statistical testing makes it difficult to rule out random fluctuation, particularly in the smaller subsets.

Conclusion: The methodology is sound, and the use of standard benchmarks (SWE-bench, SWE-QA) provides a solid foundation. However, the small size of the RL training set and the lack of variance reporting or multi-seed experiments weaken the robustness of the claims. The

paper should be accepted pending the addition of variance metrics and a brief discussion on the generalizability of the results beyond the primary GPT-5.4 configuration.

Action items.

- **[science]** The RL training set is extremely small (400 prompts, 395 repos). The paper must report variance metrics (e.g., standard deviation or confidence intervals) across multiple random seeds or bootstrap resampling to rule out that the reported gains are due to favorable sampling of easy instances.
- **[science]** The SFT data (2,954 examples) is entirely distilled from Sonnet 4.6 traces. The authors must provide evidence that the 4B model learns a generalizable exploration strategy rather than simply overfitting to the specific trajectory patterns of the teacher model, especially given the small dataset size.
- **[science]** The end-to-end evaluation relies on a single main agent configuration (GPT-5.4) for the primary cost/success claims. To support the claim of a general ‘efficient repository explorer,’ the authors should demonstrate that the subagent’s benefits hold across at least one other distinct main agent architecture or model family.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical reporting in the manuscript requires clarification to ensure the robustness of the reported gains. While the experimental design (SFT followed by RL on a held-out set) is sound, the presentation of results lacks necessary statistical context.

First, the primary results tables (Table 1: End-to-End Results; Table 2: Standalone Exploration Quality) report only point estimates (e.g., $F1 = 73.71$, $\text{Score} = 71.7$). Given the sample sizes (e.g., 200 instances for SWE-bench Pro, 300 tasks for cost analysis), the absence of variance measures (standard deviation, standard error, or confidence intervals) makes it impossible to determine if the observed improvements (e.g., the 5.5% accuracy gain) are statistically significant or within the noise of the evaluation. For instance, in the standalone evaluation, reporting the distribution of F1 scores across instances would better illustrate the consistency of the explorer’s performance compared to baselines.

Second, the aggregation method for the exploration metrics in Appendix e000 (Section “Standalone exploration evaluation protocol”) needs explicit definition. The text mentions computing per-instance precision/recall/F1 and then averaging. In information retrieval tasks with varying numbers of ground-truth citations per instance, the “macro-average” (mean of per-instance F1) and “micro-average” (F1 of aggregated counts) can differ significantly. The authors must specify which metric is reported to allow for correct interpretation and reproducibility.

Finally, regarding the RL optimization details in Appendix e000, the KL coefficient is listed as

0.0. While this simplifies the reward function, it removes the regularization term that typically prevents policy collapse in GRPO. The authors should briefly discuss the stability of the training curves (Figure 1) in this context or confirm that the reward signal was sufficiently sparse and well-defined to prevent divergence without KL regularization. The current text does not address this potential statistical risk in the training dynamics.

Action items.

- **[science]** Report variance measures (standard deviation or standard error) for all aggregate metrics (F1, Precision, Recall, Token counts) in Tables 1 and 2. Currently, only point estimates are provided, preventing assessment of statistical significance or effect stability across the 200+ instance benchmarks.
- **[science]** Clarify the statistical aggregation method for the ‘Standalone Exploration Evaluation’ (Appendix e000). The text states ‘Scoring & Compute per-instance precision... then report instance-wise averages.’ Confirm if the reported F1 is the mean of per-instance F1 scores or the F1 computed from aggregated true/false positives across all instances, as these yield different results for imbalanced data.
- **[writing]** The RL reward function (Eq 1 & 2, Appendix e000) includes a hard penalty for format violations. Specify if the reported ‘RL reward’ curves in Figure 1 are normalized or raw, and whether the KL coefficient (0.0) implies no KL-divergence penalty was applied during the reported runs, which may affect the stability of the policy updates.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript demonstrates a generally high standard of technical writing with clear, concise sentences and effective use of terminology. The logical flow from the introduction of the problem to the proposed solution and experimental validation is coherent. However, there are specific areas where the presentation of data and structural organization impede readability.

First, the “SWE-bench Pro Subset” appendix (found in chunks e002 and e003) presents a significant readability issue. The authors list 200 instance IDs in a single, unbroken enumerated list. This creates a “wall of text” that is difficult to scan and disrupts the reading flow. While the data is necessary for reproducibility, presenting it as a raw list in the main text is suboptimal. It is recommended to either move this list to a supplementary material file or format it as a compact table within the appendix to improve visual clarity.

Second, there is a minor inconsistency in the “Prompt Templates” section (e001). The text introduces the section by stating, “Below is a representative example (full texts are included in the original source).” Given that this document is the full manuscript, the phrase “original source” is ambiguous and potentially confusing to the reader. It should be clarified whether

the full texts are provided in the subsequent appendix or if the examples shown are indeed the complete templates.

Finally, in the “Standalone Exploration Evaluation Protocol” section (e001), the text references “Table~\ref{tab:explore-protocol}”. While the label exists in the source (e003), the surrounding text and the table structure in the provided chunks suggest a potential disconnect in how the table is introduced versus how it is rendered. Ensuring that the reference text explicitly describes the table’s content (e.g., “Table X details the protocol stages...”) rather than just pointing to it would enhance the narrative flow.

Overall, the writing is strong, but addressing the formatting of the large data list and clarifying the “original source” reference will significantly improve the document’s polish and readability.

Action items.

- **[writing]** In Section ‘Standalone Exploration Evaluation Protocol’ (e001), the text references ‘Table~\ref{tab:explore-protocol}’ but the subsequent table environment is labeled ‘tab:explore-protocol’ in the source (e003) yet the caption in e003 is missing the table environment start, creating a broken reference chain. Ensure the label and reference match the actual table structure.
- **[writing]** The ‘SWE-bench Pro Subset’ appendix (e002/e003) lists 200 instance IDs in a single enumerated list. This creates a massive wall of text that disrupts readability. Consider moving this list to a separate supplementary file or a compressed table format in the main text to improve flow.
- **[writing]** In the ‘Prompt Templates’ section (e001), the text states ‘Below is a representative example (full texts are included in the original source).’ Since this is the full manuscript, this phrasing is confusing. Clarify if the full texts are in the appendix or if the examples provided are the full texts.